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AIUB, Chemistry Notes.

Chemistry

FINAL TERM

GOOD LUCK

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"GOOD LUCK"

#Solutions & Solubility

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Solutions:- (দ্রবণ)

Q· What is solutions:?

⇒ A solution is a homogeneous mixture two or more pure substances.

Types of Solution:-

(1)

Q:- Classify the solution based on physical state?

⇒ Depending on the three physical states, gas, liquid, solid there are 9 types of solution.

Types	Example:
1· Gas in Gas	Air
2· Gas in liquid	Soda water
3· Gas in solid	H_2 gas absorbed by heated Palladium
4· Liquid in Gas	Water Vapor in air (জলীয়বাষ্প)
5· Liquid in Liquid	Alcohol in water
6· Liquid in Solid	mercury in gold
7· Solid in Gas	Camphor in Air (কর্পূর)
8· Solid in Liquid	Sugar Solution
9· Solid in Solid	Ni-Cu alloy

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(Q) Depending on the temperature:-

Q:- Classify the solution Based on temperature?

⇒ There are two types:-

1. Exothermic Solution

2. Endothermic Solution.

Exothermic Solution:-

⇒ An exothermic solution refers to a chemical solution that releases heat to it's surrounding. In other words, during the process of dissolving. The system loses energy in form of heat.

Ex:- CaO, NaOH, strong acid on H_2O

Endothermic Solution:-

⇒ An endothermic Solution is a chemical solution that absorbs heat from it's surrounding. In other words, during the process of dissolving the system gains Energy in form of heat.

Ex:- NH_4NO_3, $C_6H_{12}O_6$ in H_2O / evaporation of H_2O

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(3) Depending on the equilibrium:-

⇒ There are three types.

1. Saturated Solution.
2. Supersaturated Solution.
3. Unsaturated Solution.

☒ Saturated Solution:-

⇒ A saturated solution is a solution that contains the maximum amount of solute that that can be dissolved in a given solvent at a specific temperature and pressure.

→ In a Saturated Solution, any additional solute added will not dissolve and will remain as undissolved particles at the bottom of the container.

☒ SuperSaturated Solution:-

⇒ A SuperSaturated Solution is solution that contains more solute than a saturated solution.

Note:- The solubility of a solution increases as the temperature or pressure increased.

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Unsaturated Solution:-

⇒ An Unsaturated Solution is a solution containing less Solute than a Saturated solution. It has a room for more solute to be added and fully dissolved.

Solvent:- (দ্রাবক)

Q:- What is Solvent?

⇒ A Solvent is any Substance, usally Liquid, which is Capable of dissolving one or Several substance, thus creating a Solution.

Q:- What are the properties of good Solvent?

1: It should be inert to the reaction condition.

2: It should dissolve the reactants and reagents.

3: It should have an appropriates boiling point.

4: It should be easily removed at the end of the reaction.

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Polarity of a Solvent:-

⇒ There are three mesures of the polarity of a Solvent

1. Dipole moment

2. Dielectric Constant

3. Miscibility with water.

Dipole moment:-

⇒ It occurs due to equal amount of positive and negative charge separated by a distance within a molecule.

$$\boxed{\mu = q \times r}$$

μ = dipole moment unit Debye(D)

q = charge of atom unit Coulomb (C)

r = distance between charges unit Å

Ex:-

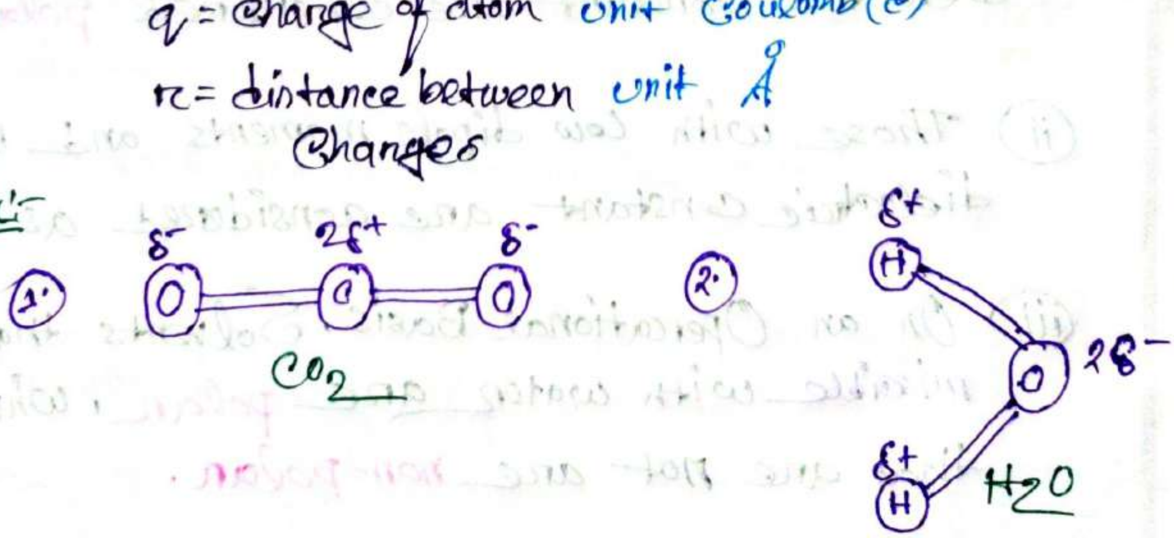

Fig:- Dipole moment

Dielectric Constant :-

⇒ The dielectric Constant is a mesure of a material's ability to reduce the electrostatic forces between charged particales, reflecting the material's polarity and it's capacity to shield.

$$\boxed{F = \frac{e^2}{rd}}$$

Where:

F = force of attraction

e = charge

r = distance between charges.

D = dielectric Constant.

Note: ① Molecules with large dipole moments and high dielectric Constant are considered polar.

(ii) Those with low dipole moments and small dielectric constant are considered as non-polar.

(iii) On an Operational Basis, Solvents that are miscible with water are polar, while those that are not are non-polar.

Solubility:- (দ্রাব্যতা)

⇒ The amount of a substance that dissolves in a given quantity of solvent at a given temperature to form a saturated solution is called its solubility.
(কোনো নির্দিষ্ট তাপমাত্রায় নির্দিষ্ট পরিমাণ দ্রাবকে সম্পৃক্ত দ্রবণে পরিণত করতে যে পরিমাণ দ্রব দ্রবীভূত হয় তাকেই ঐ তাপমাত্রায় ঐ দ্রবের দ্রাব্যতা বলে।)

Factors Affecting Solubility:-

ⓐ Nature of the solute and solvent:-

⇒ (i) Ionic and polar substances dissolve in polar solvent
Ex:- NaCl in H_2O

(ii) Non-polar Substance are dissolved in non-polar Solvent.
Ex:- Napthalene, oil etc in C_6H_6, CCl_4.

Note:- "Like Dissolves Like" - that means Substance dissolve in chemically similar solvents.

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(b) Effect of temperature:-

⇒ A higher temperature increases both the rate of disolution and also the solubility of the solute.

(1) Rate of solution:-

⇒ (i) At a low temperature the rate of solution is quite low. At higher temperature dissolution of more solute causes.

(ii) The rate of dissolution may be inereaed by shaking of the solvent-solute mixture.

(iii) Particle size factor:-

⇒ It's affects the rate of solution because the process primarily occurs at the surface. Greater surface area between solvent and solute leads to a faster rate of solution.

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Mechanism of dissolution:-

The process of dissolution of NaCl in H_2O

⇒ ⓐ H_2O molecules attract the Na^+ ions from crystal lattice, despite the attraction from nearby Cl^- ions.

ⓑ Solvent particles may need to separate to accommodate the Na^+ ions.

ⓒ The Na^+ ions are solvated by solvent molecules.

Note:- (ii) Operations ⓐ, and ⓑ require energy.
(iii) Operations ⓒ releases energy.

Q:- How do the evolution and the absorption of heat arises?

⇒ In a chemical solution, there can be either evolution or absorption of heat. Based on this the solution can be two types.

1. Endothermic Solution:- The Solution in which heat is absorbed.

2. Exothermic Solution:- The solution in which heat is released.

⇒ when the energy required for the operations ⓐ and ⓑ is greater than that released from operation ⓒ, the temperature of a system goes down. that means heat is absorbed during the dissolution of NaCl molecule. The amount of heat absorbed is called positive heat of solution (endothermic System).

⇒ When the energy required for the operations ⓐ and ⓑ is less than that released from operation ⓒ, the temperature of a system goes up that means heat is evolved during the dissolution of NaCl molecule. The ammount of heat evolved is called negative heat of solution (exothermic System)

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problem : 1 :-

=> Calculation of Normality of strong acids.

(a) 36% (w/w) HCl, specific gravity 1.18

(b) 96% (w/w) H_2SO_4, specific gravity 1.84

(a)

=> we know:-

$$\text{Normality (N)} = \frac{\text{Percentage by weight} \times S \times 1000}{\text{Gram equivalent weight}} \quad \text{--- (1)}$$

=> is given:-

Percentange by weight = 36% (w/w)

$S = 1.18$

$$G.E.W = \frac{36.5}{1} = 36.5g$$

put the value in eq (1) =>

$$N = \frac{36\% \times 1.18 \times 1000}{36.5}$$

$$= 11.638N$$

(b)

=> is given:-

Percentange by weight = 96% (w/w)

$S = 1.84$

$$G.E.W = \frac{98}{2} = 49g$$

put the value in eq (1) =>

$$N = \frac{96\% \times 1.84 \times 1000}{49} = 36.048 \ N$$

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Gas Laws:-

⇒ The gas laws are a set of laws that describe the relationship between temperature (T), pressure (P) and volume (V) of gases.

Boyle's Law:-

⇒ Boyle's Law state that the product of pressure and volume of a fixed quantity of an ideal gas is constant, given constant temperature.

$$\boxed{PV = K}$$

$$\boxed{P_1V_1 = P_2V_2}$$

<u>where,</u>

P = pressure

V = volume

K = Constant.

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#Charle's law

⇒ charles law state that the volume of an ideal gas is directly proportional to the absolute temperature at constant pressure.

$$\boxed{V \propto T}$$

$$\Rightarrow \boxed{\frac{V}{T} = K}$$

$$\therefore \boxed{\frac{V_1}{T_1} = \frac{V_2}{T_2}}$$

Where,

V = volume

T = Temperature

K = Constant.

#Gay-Lussac's Law:-

⇒ It state that pressure of a fixed amount of gas at fixed volume is directly proportional to it's temperature.

$$\boxed{P \propto T}$$

$$\boxed{\frac{P}{T} = K}$$

$$\boxed{\frac{P_1}{T_1} = \frac{P_2}{T_2}}$$

Combined gas law:-

⇒ These three laws were combined to form the combined gas law.

$$\boxed{\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}}$$

Ideal gas:-

⇒ Ideal gases are hypothetical gases that maintain the ideal gas law under all conditions of temperature and pressure.

$$\boxed{PV = nRT}$$

where:- P = pressure (SI Units: Pascal (Pa))
V = volume (SI Units: Cubic meter (m^3))
n = No. of moles
R = 8.3145 J/(mol K)
T = Temperature (SI Units: Kelvin (K))

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Avogadro's Law:

⇒ It state that under the same conditions of temperature and pressure, equal volumes of different gases contain an equal number of molecules.

∴ Avogadro's no = 6.023×10^{23}

Note:- The law, often called Avogadro's hypothesis is true only for ideal gases.

S.T.P

⇒ S.T.P = Standard Temperature and pressure are used when comparing the properties of gases.

T = 273K (0°C)

P = 101325 Pa (1 atm)(760 mmHg)

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Solubility Product & pH

Chaos
Now

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Solubility product

Q. What is solubility product law with example?

⇒ The product of the concentration of the ions that exist in equilibrium with the solid compound in a saturated solution.
(একটি স্যাচুরেটেড দ্রবণে কঠিন যৌগের সাথে সাম্যাবস্থায় বিদ্যমান আয়নগুলির ঘনত্বের গুণফল)

Example:-

$$\underset{\text{solid}}{PbSO_4} \rightarrow \underset{\text{in solution}}{Pb^{2+} + SO_4^{2-}}$$

Now, According to the Solubility product law

$$\boxed{[Pb^{+2}][SO_4^{-2}] = \text{a Constant.}}$$

where:-

$[Pb^{+2}]$ = Concentration of Pb^{+2} ions as gram ions per liter

$[SO_4^{-2}]$ = Concentration of SO_4^{-2} ions as gram ions per liter.

problem 1:-

⇒ The solubility of $PbSO_4$ in H_2O at 25°C is found to be 0.0037 gm/100 gm H_2O. Find its solubility product (K_{sp}) at that temperature (mol. wt. $PbSO_4$ is 303.37)

⇒ Is given:-

The molecular weight of $PbSO_4$ = 303.37

We know:-

1L = 1000mL

1mL = 0.001L

For pure water 1mL = 1g

∴ The solubility of $PbSO_4$ = $\frac{0.0037\,gm}{100\,gm}$ H_2O

$= \frac{0.0037\,gm}{100\,mL}$ H_2O

$= \frac{0.0037\,gm}{100\,mL \times 0.001L}$ H_2O

$= 0.037\,gmL^{-1}$ H_2O

So, $= \frac{0.037}{303.37}$ mol/L H_2O

$= 1.21 \times 10^{-4}$ mol/L H_2O

Since, $PbSO_4$ dissociates completely in H_2O.

So, Each molecule of it on dissociation productes.

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$\therefore [Pb^{+2}] = 1.2\times10^{-4}$ moles ions per liter

$[SO_4^{-2}] = 1.2\times10^{-4}$ moles ions/liter

So $K_{sp} = [Pb^{+2}] * [SO_4^{-2}]$

$= [(1.2\times10^{-4})\times(1.2\times10^{-4})]$

$= 1.44\times10^{-8}$

Ans:-

problem: 2

⇒ The solubility product of $CuCl_2$ is 3.2×10^{-7} at 25°C. Calculate the solubility of $CuCl_2$ in mole liter^{-1}

⇒ $CuCl_2$ is a sparcingly soluble salt.

Let,

Solubility of $CuCl_2 = x$ in moleL^{-1}

Now:-

$$CuCl_2 \rightleftharpoons Cu^{+2} + 2Cl^-$$

Equilibrium Concentration $\quad x \quad x \quad 2x$

is given:-

$K_{sp} = 3.2\times10^{-7}$

We know:-

$K_{sp} = [Cu^{+2}]\times[Cl^-]^2$

$K_{sp} = x\times(2x)^2$

$K_{sp} = 4x^3$

$\therefore x = \sqrt[3]{\frac{K_{sp}}{4}} = \sqrt[3]{\frac{3.2\times10^{-7}}{4}} = \boxed{4.3\times10^{-3}}$ mole/liter

Ans:-

problem: 3:-

=> K_{sp} of CaF_2 is 1.7×10^{-10} and its mol. wt. is 78 g $mole^{-1}$. What volume of the saturated solution will contain 0.078g of CaF_2?

=> In given:-

$K_{sp} = 1.7\times10^{-10}$

m = mass of substance = 0.078g

M = Mass of one mole = 78 g$mole^{-1}$

We know:-

No. of mole $n = \frac{m}{M}$

$= \frac{0.078}{78}$ mole

$= 1.0\times10^{-3}$ mole

Let,

Solubility of $CaF_2 = x$ in $mole L^{-1}$

Now!

$$CaF_2 \rightleftharpoons Ca^{2+} + 2F^-$$

∴ Equilibrium Concentration $\quad x \quad x \quad 2x$

We know:-

$K_{sp} = [Ca^{2+}][F^-]^2$

$= x\times(2x)^2$

$= 4x^3$

$\therefore x = \sqrt[3]{\frac{1.7\times10^{-10}}{4}} = 3.480\times10^{-4}\ mol\ L^{-1}$

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$\therefore V = \frac{n}{C} L$

$= \frac{1.0 \times 10^{-3}}{3.480 \times 10^{-4}} L$

$= 2.86 L$

$\therefore$ 0.078g of CaF_2 in Contained in 2.86L of the saturated Solution.

Ans:-

Solubility Product VS Ionic Product

⇒ The Solubility product of an insoluble substance is the product of the Concentrations of its ions at equilibrium.

⇒ The ionic product is the product of actual Concentration of it's ions that may or may not be equilibrium with the solid.

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Solubility Product Principle:-

1. When the ionic product is $\boxed{\text{equal}}$ to the Solubility product, the solution is $\boxed{\text{Saturated}}$

2. when the ionic product is $\boxed{\text{greater}}$ than the Solubility product, the solution is $\boxed{\text{Supersaturated.}}$

3. when the ionic product is $\boxed{\text{less}}$ than the Solubility product, the solution is $\boxed{\text{unsaturated}}$

Common Ion Effect:-

⇒ The shift in equilibrium that occurs because of the addition of an ion, which already involved in the equilibrium reaction.

$$AgCl(s) \rightleftharpoons Ag^{(+)} + Cl^{(-)}$$

adding NaCl(aq) shift equilibrium position.

→ Adding of NaCl shift equilibrium to left due to excess of Cl^- ions and decrease solubility of AgCl.

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Application of Solubility product principle:-

Purification of common salt:-

⇒ A saturated solution of common salt is prepared and insoluble impurities are filtered off. Ane HCl gas passed through it. The equilibrium:-

$$NaCl \rightleftharpoons Na^+ + Cl^-$$
$$HCl \rightleftharpoons H^+ + Cl^-$$

The concentration of Cl^- ions increases in the solution due to ionization HCl. Hence, the ionic product $[Na^+][Cl^-]$ exceeds the solubility product. The result is a supersaturated solution of NaCl, therefore pure NaCl precipitates out from solution.

Salting out of soap:-

⇒ Soap is a sodium salt of higher acids. From the solution, soap is precipitated by the addition of concentrated solution of NaCl. They present in the form of ions.

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$$C_nH_{2n+1}COONa \rightleftharpoons C_nH_{2n+1}COO^- + Na^+$$

Soap

$$NaCl \rightleftharpoons Na^+ + Cl^-$$

thus, the Concentration of Na^+ ions increases on addition of NaCl solution. Hence the ionic product $[C_nH_{2n+1}COO^-][Na^+]$ exceeds the solubility product of soap. The result is a SuperSaturated solution of soap. Therefore, soap precipitates out from the solution.

problem:4:-

⇒ Calculate the solubility of AgCl ($K_{sp} = 1.7\times10^{-10}$) in 0.01M NaCl Solutions

⇒ In given:-

K_{sp} of $AgCl = 1.7\times10^{-10}$

Let, Solubility of $AgCl = x$ in M

Now:-

$$AgCl \rightleftharpoons Ag^+ + Cl^- \quad \& \quad NaCl \rightleftharpoons Na^+ + Cl^-$$

∴ equilib. conc: $x \quad x \quad x$. $0.01 \quad 0.01 \quad 0.01M$

∴ Total Concentration of Cl^- in soln $= 0.01M + xM$

Since:- AgCl is sparingly soluble,

So x is negligibly small.

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So, $[Cl^-] = 0.01M$

We know:-

$$K_{sp} = [Ag^+] \times [Cl^-]$$

$$1.7 \times 10^{-10} = [x] \times [0.01]$$

$$\therefore x = \frac{1.7 \times 10^{-10}}{0.01}$$

$$= 1.7 \times 10^{-8} M$$

$\therefore$ The solubility of AgCl in 0.1M NaCl solution is $1.7 \times 10^{-8} M$

Ans:-

problem: 5:-

⇒ Ksp of $Mg(OH)_2$ is 1.8×10^{-11} at 25°C. Calculate the solubility of $Mg(OH)_2$ in 0.1M aqueous NaOH solution.

⇒ in given:-

$$K_{sp} \text{ of } Mg(OH)_2 = 1.8 \times 10^{-11}$$

Let, Solubility of $Mg(OH)_2 = x$ in M

Now,

$$Mg(OH)_2 \rightleftharpoons Mg^{+2} + 2OH^- \quad \& \quad NaOH \rightleftharpoons Na^+ + OH^-$$

Equilib. Conc:- x x $2x$ 0.1 0.1 $0.1M$

$\therefore$ Total Concentration of OH^- in the solution

$= 0.1M$ (from NaOH) $+ 2x\,M$ (from $Mg(OH)_2$)

Since, $Mg(OH)_2$ is sparingly soluble.

So, x is negligibly small.

$$\therefore [OH^-] = (0.1M + 2x\,M) \cong 0.1M$$

We know:-

$$K_{sp} = [OH^-]^2[Mg^{2+}]$$

$$K_{sp} = (0.1)^2 \times x$$

$$\therefore x = \frac{1.8\times10^{-11}}{(0.1)^2}$$

$$= 1.8\times10^{-9}\,M$$

Ans:-

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Ionic product of water:-

পানির আয়নিক গুণফল

⇒ Water is known to be slightly ionized.

$$H_2O \rightleftharpoons H^+ + OH^-$$

But water can both gain H^+ ions to from H_3O^+ (acting as a base) and donate H^+ ions (Acting as a acid)

$$H_2O + H_2O \rightleftharpoons H_3O^+ + OH^-$$

Sol: Applying law of equilibrium,

$$K = \frac{[OH^-]\times[H^+]}{[H_2O]}$$

Now:- In case of dilute Solution:-

$$K \times [H_2O] = [H^+]\times[OH^-]$$

Here, K_w = ionic product of water is the constant product of H^+ and OH^- ion concentration in pure water at 25°C temperature.

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$\therefore K_w = 1.0 \times 10^{-14}$

⇒ In pure water and neutral solution, the molar Concentration of H^+ ions and OH^- ions are equal.

$\therefore [H^+] = [OH^-]$

Since $[H^+] \times [OH^-] = 1.0 \times 10^{-14}$ ....

So, $[H^+] \times [H^+] = 1.0 \times 10^{-14}$

$[H^+]^2 = 1.0 \times 10^{-14}$

$\therefore [H^+] = \sqrt{1.0 \times 10^{-14}}$

$= 1 \times 10^{-7}$ mol/L

$\therefore [OH^-] = 1 \times 10^{-7}$ mol/L $[\because [H^+] = [OH^-]]$

⇒ Now The concentration of H^+ ions and OH^- ions can be expressed in terms of the ionic product of water (Kw) as follows:-

$$\boxed{[H^+] = \frac{K_w}{[OH^-]} = \frac{1 \times 10^{-14}}{[OH^-]}}$$

$$\boxed{[OH^-] = \frac{K_w}{[H^+]} = \frac{1 \times 10^{-14}}{[H^+]}}$$

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Note:-

(i) For neutral solution $[H^+] = [OH^-] = \sqrt{(1\times10^{-14})} = 1\times10^{-7}$ mol/L

(ii) For acidic solution $[H^+] > 1\times10^{-7} > [OH^-]$ mol/L

(iii) For Basic Solution $[OH^-] > 1\times10^{-7} > [H^+]$ mol/L

pH Value:-

Q:- What is pH?

⇒ pH is defined as the negative logarithm of the concentration of hydrogen ions $([H^+])$ in mol/L.

The pH scale typically ranges from 0-14

pH - Value of $[H^+]$ & $[OH^-]$

$$\boxed{pH = -\log_{10}[H^+]}$$

$$\boxed{pOH = -\log_{10}[OH^-]}$$

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we know:-

For pure water,

$$[H^+]\cdot[OH^-] = 1\times10^{-7}\ mol/L$$

$$\therefore pH = -\log[H^+] = 7$$

$$\therefore pOH = -\log[OH^-] = 7$$

$$\therefore \boxed{pH + pOH = 7+7 = 14}$$

problems 2:-

⇒ Calculate the pH of 0.001 M HCl

⇒ HCl is a strong acid. it is completely dissociated in aq solution.

$$HCl \rightleftharpoons H^+ + Cl^-$$

$$0.001M \quad 0.001M \quad 0.001M$$

we know:-

$$pH = -\log[H^+]$$

$$= -\log(0.001)$$

$$= 3$$

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problems6:-

⇒ Calculate pH and pOH of 0.02M H_2SO_4 solution. $K_w = 1\times10^{-14}$ at 25°C

⇒ Here,

$$H_2SO_4 \rightleftharpoons 2H^+ + SO_4^{-}$$

Equilib Conc. x $2x$ x

is given:-

$x = 0.02M$

So, $[H^+] = 2\times0.02 = 0.04M$

we know:-

$$pH = -\log[H^+]$$
$$= -\log[0.04]$$
$$= 1.397$$

$$\therefore pOH = 14 - pH$$
$$= 14 - 1.397$$
$$= 12.60$$

Another way:-

$$[OH^-] = \frac{K_w}{[H^+]} = \frac{1\times10^{-14}}{0.04} = 2.5\times10^{-13}M$$

$$\therefore pOH = -\log[OH^-] = -\log(2.5\times10^{-13}) = 12.60$$

Ans:-

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problem: 7:-

⇒ pH of an aq. solution of HCl is 2.699 at 25°C Calculate the molarity of solution.

⇒ In given:-

$$pH = 2.699$$

Now:

Let,

Molarity of solution = x

$$\underset{x}{HCl} \rightleftharpoons \underset{x}{H^+} + \underset{x}{Cl^-}$$

we know:-

$$pH = -\log[H^+]$$

$$\therefore x = \text{antilog}(pH)$$

$$= \text{antilog}(-2.699)$$

So, $x = 1.99 \times 10^{-3} M$

Ans:-

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Importance of pH in our Daily Life:-

1. $\boxed{\text{Our body works}}$ within the pH rang of $\boxed{7.0 \text{ to } 7.8}$

2. $\boxed{\text{PH of rain}}$ water is $\boxed{\text{less than } 5.6}$. called Acid rain.

3. $\boxed{\text{Tooth decay}}$ starts when pH of the mouth is $\boxed{\text{Lower}}$ than $\boxed{5.5}$

4. $\boxed{\text{Our Blood}}$ pH range $\boxed{7.35 \text{ and } 7.45}$

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#Electro Chemistry :-

Ch. 02

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Electro-Chemistry:-

⇒ is the study of electron movement in an oxidation or reduction reaction at a polarized electron surface.

Electrolytes:-

⇒ Electrolytes are electrovalent substance that form ions in solution which conduct an electric current. (এটি হলো এমন একটি পদার্থ যা দ্রবণে আয়ন তৈরি করে এবং ইলেকট্রিসিটি প্রবাহ করে)

Electrolysis.

⇒ It's a process, which electric current is passed through a substance to effect a chemical change

OR

⇒ The process of decomposition of an electrolyte by passing electric current through its solution is called electrolysis.

(একটি ইলেক্ট্রোলাইট দ্রবণের মধ্যে দিয়ে বিদ্যুৎ প্রবাহের মাধ্যমে তার পচন ঘটানোকে ইলেকট্রোলাইসিস বলে।

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Note:-

(i) It's a method of using a direct electric (DC) current.

(ii) It's commercially important for purification and separation.

iii It's also used for the protection and beautification of metalic materials known as electroplating.

Electrical Units:-

1. Colump → A unit quantity of electricity

2. Amphere → A unit rate of flow of electricity

3. Ohm → A unit of electric resistance(R)

4. Volt → A unit of electromotive force(V)

Conductance of electrolytes:-

⇒ The ability of an electrolyte to conduct an electric current is called conductivity OR Conductance of electrolyte. denote by C

$\therefore \boxed{C = R^{-1}}$ ohm^{-1} or Siemens (s)

Where,

C = Conductance

R = Resistance.

Types of Conductance:-

⇒ There are two types of Conductance.

1. Specific - Conductance (K, kappa)
2. Equivalent - Conductane (λ, lambda)

Specific - Conductance (k, kappa)

⇒ The Conductance of one centimeter cube of a solution of an electrolyte is called 'specific Conductance'. denote by 'K'.

$$\boxed{K = \frac{1}{\rho} = \frac{1}{R} \times \frac{l}{A} = \frac{l}{RA}}$$ ohm^{-1}cm^{-1}

$\therefore R \propto \frac{1}{A}$ or $R = \rho \times \frac{1}{A}$

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Equivalent - Conductance:-

⇒ The conductance of a volume of solution containing one equivalent of an electrolyte. denote by λ

$$\boxed{\lambda = K \times V}$$

$$\boxed{\lambda = \frac{1}{R} \times \frac{l}{A} \times V} = \frac{1}{ohm} \times \frac{cm}{cm^2} \times \frac{cm^3}{eqvt} = ohm^{-1} cm^2 eqvt^{-1}$$

Kohlrauch's Law:-

⇒ The equivalent conductance of an electrolyte at infinite dilution* ($\lambda\alpha$) is equal to the sum of the equivalent conductance of the component ions

$$\boxed{\lambda\alpha = \lambda a + \lambda c}$$

λa = equivalent Conductance of the anion

λc = equivalent Conductance of the Cation.

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Example:-

⇒ The equivalent conductance of NaCl at infinite dilution at 25°C is 126.45. The equivalent conductance of Na^+ and Cl^- ions are 50.11 ohm^{-1} and 76.34 ohm^{-1}

⇒ So, we know:-

$$\lambda(NaCl)_\infty = \lambda(Na^+)_\infty + \lambda(Cl^-)_\infty$$

$$126.45 = 50.11 + 76.34$$

$$\therefore 126.45 = 126.45$$

problem 1:-

⇒ Calculate the equivalent conductance of NH_4OH at infinite dilution at 20°C is $\lambda_\infty(NH_4Cl) = 130$, $\lambda_\infty(OH^-) = 174$, $\lambda_\infty(Cl^-) = 66$.

⇒ In given:-

$$\lambda_\infty(NH_4Cl) = 130$$

$$\lambda_\infty(OH^-) = 174$$

$$\lambda_\infty(Cl^-) = 66$$

we know:-

$$\lambda_\infty(NH_4OH) = \lambda_\infty(NH_4Cl) + \lambda_\infty(OH^-) - \lambda_\infty(Cl^-)$$

$$= 130 + 174 - 66$$

$$= 238 \ ohm^{-1} cm^2 eqv.t^{-1}$$

Ans:-

Conductometric Titrations:-

⇒ It's a method of quantitative analysis used to identify the concentration of a given analyte in a mixture.

Titration of a strong acid (HCl) against a strong base (NaOH)

Here,

$HCl \longrightarrow H^+ + Cl^-$

$H^+ + Cl^- + Na^+ + OH^- \longrightarrow Na^+ + Cl^- + H_2O$

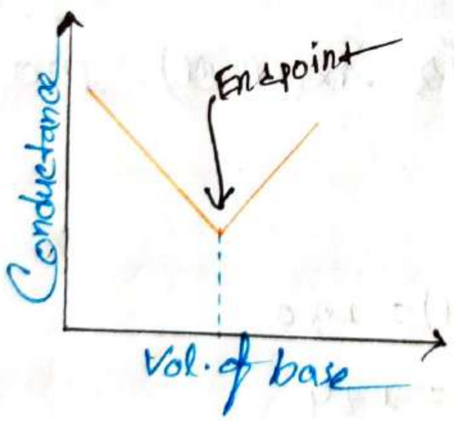

Fig:- Conductometric titration Curve For HCl and NaOH

1. Before the alkali is added, the conductivity of the solution is due to the presence of H^+ and Cl^- ions. (ক্ষার যুক্ত করার আগে, H^+ এবং Cl^- আয়নগুলোর উপস্থিতির কারণে দ্রবণ পরিবাহিত হয়)

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(2) H^+ ion has the highest mobility of any ion, and that's why the conductane is high.

(3) After adding NaOH, H^+ is removed by OH^- forming weakly ionized H_2O molecules. and the place of H^+ ions slowly taken by Na^+ ions.

(4) As more NaOH is added, the conductane of the solution decreases and reaches a minimum point (end-point)

(5) As more NaOH is added, the conductane increases since the OH^- ions are not consumed in the reaction in form of H_2O molecules.

(end-point এ পৌঁছানোর পরে যদি NaOH দ্রব দ্রবণ আরও যোগ করা হয় হয়, তাহলে আয়ন পরিবাহিতা বৃদ্ধি পায় কেননা OH^- আয়নগুলো H_2O অণু আকারে বিক্রিয়ায় consumed হয় না).

Titration of weak acid (CH_3COOH) Against a strong Base ($NaOH$)

Note

$CH_3COOH \rightarrow CH_3COO^- + H^+$

$CH_3COO^- + H^+ + Na^+ + OH^- \rightarrow CH_3COO^- + Na^+ + H_2O$

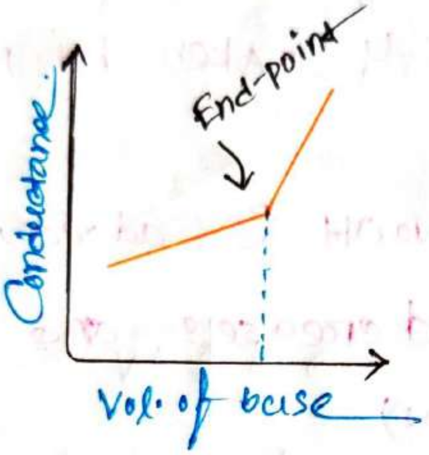

Fig:- Conductometric titration for CH_3-COOH and NaOH

1. Initial Conductance is low due to the poor dissociation of weak acid.

2. CH_3COONa formed, At first Suppress ionization due to the common-ion-effect.

4. As more NaOH is added the Conductance Starts to increase.

Titration of strong acid (HCl) against a weak base (NH_4OH)

Now:

$HCl \rightarrow H^+ + Cl^-$

$NH_4^+ + OH^- + H^+ + Cl^- \rightarrow NH_4^+ + Cl^- + H_2O$

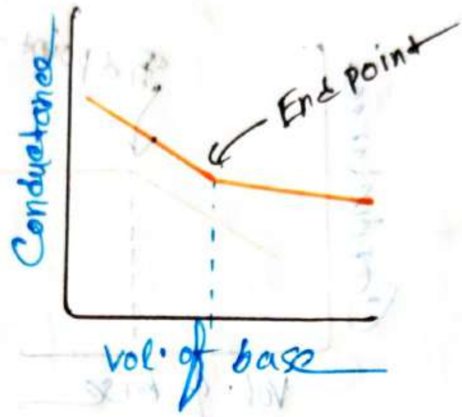

Fig: Conductrometric titration curve for HCl and NH_4OH

⇒ In this case the conductance of the solution will first decreases due to the fixing up of the fast moving H^+ ions and there replacement by slow moving NH_4^+ ions.

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Titration of weak acid (CH_3COOH) Against a weak base (NH_4OH)

NOW: $CH_3COOH \rightarrow CH_3COO^- + H^+$

$\therefore NH_4OH + CH_3COOH \rightarrow CH_3COO^- + NH_4^+ + H_2O$

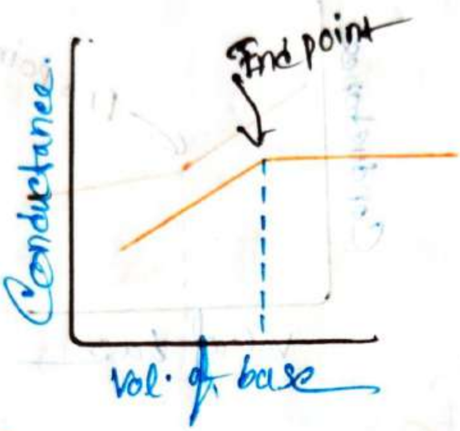

Fig: Conductometric titration Curve for CH_3-$COOH$ and NH_4OH

1. Initial Conductance is low due to the dissociation of weak acid.

2. It starts increasing as the salt CH_3COONH_4 is formed.

3. After the end-point, the conductivity remains almost constant due to the free base NH_4OH is a weak electrolyte.

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Advantages of Conductometric titration over the Volumetric titration:—

1. Colored solution, where no indicator is found to work effectively can be successfully titrated by this method.

2. The method is useful for the titration of weak acids and weak bases, where regular indicators don't show a clear color change.

3. The method gives more accurate result because we find the end-point by looking at a graph.

Differences between Conductometric titration and Volumetric titration:-

Conductometric titration	Volumetric titration.
1. Conductance mesurement are done to check end-point	1. Volume mesurement are done to check end-point
2. Titration Canbe carried Out even with ~~out~~ the colored solution	2. Titration may fail in colored solution due to the unavailability of suitable indicators.
3. Accurate result obtained	3. Result are not so Accurate.
4. End-point are determined graphically	4. End-point are determined by change of color of indicator.
5. Succeful even in weak acids and weak base	6. Not Successful in weak acid and base.

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Electrolytic and Electronic Conductors:-

Electrolytic	Electronic or metallic
=> Electrolytic Conduction occurs via the movement of ions towards opposite electrodes.	=> Electrolytic ~~Condition~~ conduction involves ions moving towards opposite electrodes
=> It appears with chemical changes at the electrode.	=> No chemical changes occur during electronic conduction
=> It become more conductive at higher temperatures	=> The resistance of these Conductors increases at higher temperature.
=> Example of electrolytic Conductors include fused Salts, Solution of Certain compounds in water. Polar Solvent	=> Metals, alloys, Carbon, and Certain solid salts like Cadmium, sulphide, Cupric sulphide are all Electronic Conductors

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Battery in Chemistry:-

Q:- What is Battery?

⇒ A battery is a device that converts chemical energy into electrical energy.

Types of Battery:-

⇒ There are two types

1. primary batteries
2. Secondary batteries.

primary Batteries:-

⇒ It can provide only - one Continuos discharge. Can not be re-charged.

The discharge reaction:-

Anode (Oxidation):- $Zn(s) \longrightarrow Zn^{2+} + 2e^-$

Cathode (Reduction):- $2NH_4^+ + 2e^- \rightarrow 2NH_3 + 2H_2$

Cell Reaction: $Zn(s) \,|\, Zn^{2+}(aq) \,||\, 2NH_4^+(aq), [2NH_3(aq) + H_2(g)] \,|\, C(gr)$

$E = 1.5V$

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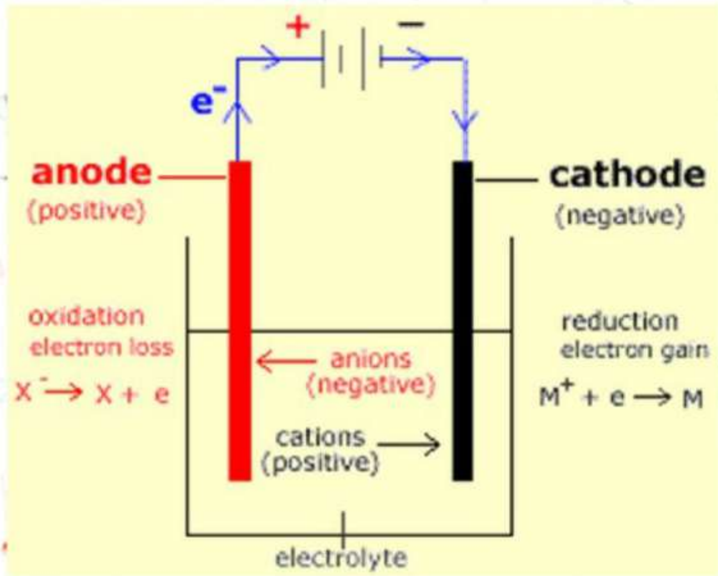
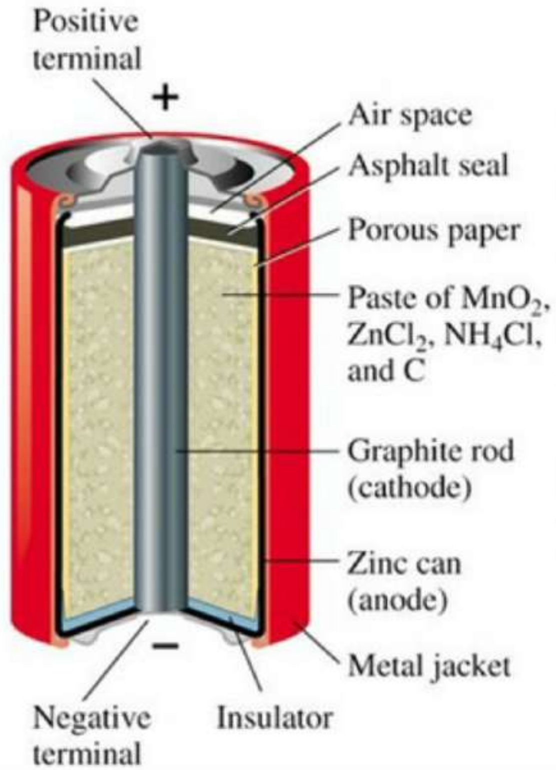
⇒ Then MnO_2 prevents H_2 from collecting on graphite rod:

$$H_2(g) + 2MnO_2(s) \longrightarrow 2MnO(OH)(s)$$

⇒ At the anode, NH_3 combines with Zn^{2+} to remove it from the reaction

$$Zn^{2+} + 4NH_3 \longrightarrow Zn(NH_3)_4^{2+}$$





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Secondary batteries:-

⇒ A secondary or storage battery is made of several chemical and elemental materials. These materials change during charging and discharging and this change reversible.

The discharging and charging reactions:-

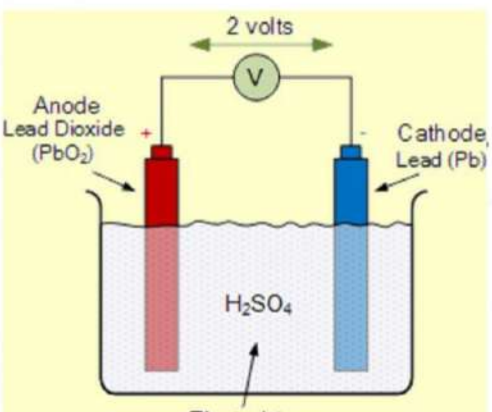
Anode (oxidation): $Pb(s) + H_2SO_4(aq) \rightarrow PbSO_4(s) + 2e^- + 2H^+(aq)$

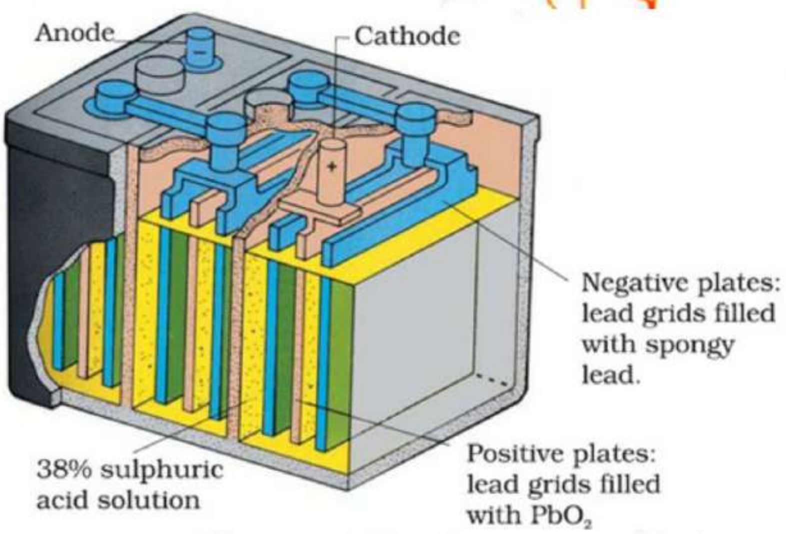
Cathode (reduction): $PbO_2(s) + 2H^+ + 2H_2SO_4(aq) + 2e^- \rightarrow PbSO_4(s) + 2H_2O(l)$

→ (discharging)

Overall Reaction:- $PbO_2(s) + Pb(s) + 2H_2SO_4(aq) = 2PbSO_4(s) + 2H_2O(l)$

← charging





⇒ Repeated charging leads to the hydrolysis of H_2O into H_2 and O_2 so distilled water in added sometimes to keep H_2SO_4 Concentrain Constant.

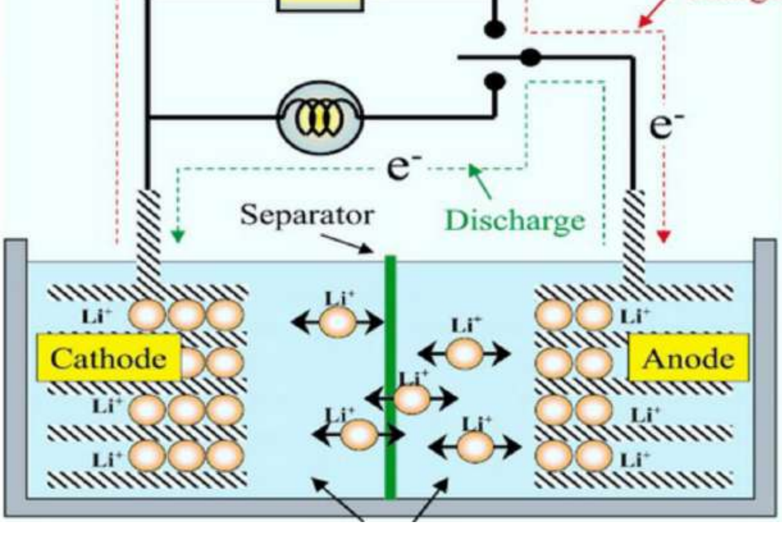
#Lithum ION Battery:-

⇒ A Li-ion battery is a rechangeble battery where Li^+ ions move between electrodes through an electrolyte during dischange and rechange.

positive electrode $\quad LiCoO_2 \underset{d}{\overset{c}{\rightleftharpoons}} Li_{1-x}CoO_2 + xLi^+ + xe^-$

Negative electrode $\quad C_n + xLi^+ + xe^- \underset{d}{\overset{c}{\rightleftharpoons}} C_nLi_x$

Overall $\quad LiCoO_2 + C_n \underset{d}{\overset{c}{\rightleftharpoons}} C_nLi_x + Li_{1-x}CoO_2$



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photolithography:-

1. Lithography: Printing method from stone or metal plate.

2. Originally for cheap theatrical publishing.

3. Mondern lithography uses polymer-coated aluminum plates.

4. photolithography:- Uses photographic images for printing.

5. Crucial for IC fabrication in microelectronics.
6. Involves light exposure through a mask.
7. Silicon wafers used, coated photoresist compounds
8. Photoresists contain solvent, resin, sensitizer.
9. processing steps: substrate prep, spin coat, exposure, development.

10. Final step: resist strip to transfer pattern.

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THERMO CHEMISTRY

Ch 8.10

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Thermochemistry:-

⇒ It is the study of the heat energy which is associated with chemical reactions and/or phase changes such as melting and boiling.

The laws of thermo-chemistry:-

⇒ there are three Laws:-

☒ First law:-

⇒ Energy is conserved, it can be neither create or destroyed.

☒ Second-law:-

⇒ In an isolated system, natural processes are spontaneous when they lead to an increase in disorder, or entropy.

☒ Third-Law:-

⇒ The entropy of a perfect crystal in zero when the temperature of the crystal is equal to absolute zero (0K)

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Energy and It's Units:-

⇒ The Ability or Capacity to do work.

E = Kinetic energy + Potential Energy

$$\boxed{E = E_K + E_p}$$

Kinetic-Energy:-

⇒ It is a property of moving object or particle depends not only its motion but also on its mass. denote by E_K

$$\boxed{\therefore E_K = \frac{1}{2}mv^2}$$ kgm^2/s^2 or Joule (J)

potential Energy:-

⇒ It is the energy that stored in an object due to its Configuration.

$$\boxed{E_p = mgh}$$ kgm^3

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⇒ The sum of the E_p and E_k is reffered to as the internal energy, U, of the substance.

$$\boxed{\therefore E_{tot} = E_k + E_p + U}$$

Law of Conservative of energy:-

⇒ Energy may be converted from one form to another but quantity of total energy remains constant.

Enthalpy and Entropy:-

⇒ Enthalpy is the central factor in thermodynamic. Every substance contains stored chemical energy called enthalpy. Enthalpy change denote by ΔH

$$\boxed{\therefore \Delta H = \Delta E + P\Delta V}$$

ΔH = Change in enthalpy

$P\Delta V$ = work done

ΔE = Internal Energy.

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⇒ Entropy is the degree of disorder or Uncertainity in the system. denoted by ΔS

$$\boxed{\therefore\ \Delta S = \int \frac{dQ_{rev}}{T}}\ J/K$$

ΔS = change in entropy

Note:-

(i) A disordered system has high entropy.

(ii) An Ordered System has low entropy.

System:-

⇒ The portion of the physical universe, which is under thermodynamic consideration is called System.

Types of System:-

⇒ there are three types.

(i) Open System : energy & matter can transfer

(ii) Closed system: energy transfer only

(iii) Isolated System: No transfer.

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Boundary:-

⇒ The real or imaginary line that marks the limit of of the system to a definite place in space is called boundary.

Note:- The Boundary separates the system from the rest of the Universe.

Surrounding:-

⇒ The portion of the physical universe outside the boundary of the system is known as surrounding.

Universe:-

⇒ It consists of a system and a surrounding of that system in space.

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#Exothermic & Endothermic System:-

1. Energy is release from the system in Exothermic System.

2. Energy is Absorbed by the system in Endothermic system.

Total Energy of a body:-

⇒ All types of energy that a body possessed are known as total Energy

there are main two types:-

① The external Energy

② The internal Energy.

each of these two can also have two sub type:-

1. Potential Energy

2. Kinetic Energy.

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External potential Energy:-

⇒ It determined by the position of an object relative to the earths surface.

External Kinetic Energy:-

⇒ It determined by the speed of the movement of an object.

Internal potential Energy:-

⇒ It refers to the energy stored within a system due to the relative position and configurations of the subatomic particles or components.

Internal Kinetic Energy:-

⇒ It refers to the energy associated with the random motion of particles within a system.

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Heat of Reaction:-

⇒ It is the amount of heat energy released or absorbed during a chemical reaction. At constant ~~tempera~~ pressure. $\boxed{\text{It also known as enthalpy}}$

Ex:-

(i) $C(s) + O_2(g) \rightarrow CO_2(g)$; $\Delta H = -93.03$ kcal

(ii) $2C(s) + 2H_2(g) \rightarrow C_2H_4$; $\Delta H = +12.5$ kcal

Note:-

(i) $\Delta H = -ve$, Exothermic

(ii) $\Delta H = +ve$, Endothermic.

Types of heat of Reaction:-

1. Heat of Combustion.
2. Heat of Formation.
3. Heat of Solution.
4. Heat of Neutralization.

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#Heat of Combustion:-

⇒ It is defined as the change of enthalpy of a System, when one mole of the substance is fully burnt in excess of air or oxygen. It always negative

Ex:-

$$CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O(l) \; ; \; \Delta H_c = -21.0 \text{ Kcal}$$

#Heat of Formation:-

⇒ The change in enthalpy that takes place when one mole of the compound is formed from its element.

$$2C(s) + H_2(g) \rightarrow C_2H_2(g) \; ; \; \Delta H_f = +53.14 \text{ KCal}$$

#Heat of Solution:-

⇒ The change of enthalpy when one mole of substance is dissolved in a specified quantity of solvent at a given temperature.

$$KCl(s) + H_2O(l) \rightarrow KCl(aq) \; ; \; \Delta H_s = -4.4 \text{ Kcal}$$

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Heat of Neutralization:-

⇒ It is defined as the change in heat content of the system when 1gm equivalent of an acid is neutralized 1gm equivalent of a base and vice versa in dilute solution.

$$NaOH_{(aq)} + HCl_{(aq)} \rightarrow NaCl_{(aq)} + H_2O_{(l)} \; ; \; \Delta H_n = -13.68 \text{ kcal}$$

Fuels:-

⇒ A fuel is any substance that is burned or similarly reacted to provide heat and other forms of energy.

Food as Fuels:-

1. Food serve three primary functions in the body:
   (i) Tissue growth and repair.
   (ii) Synthesis of regulatory compounds.
   (iii) Energy provision.

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(2) About 80% of energy derived from the food is utilized for heat production.

(3) The remaining energy is used for muscular action, chemical processes, and othe body function.

4. Energy generation from food follows. Heat of combustion method:

$$C_6H_{12}O_6 + 6O_2 \longrightarrow 6CO_2 + 6H_2O + \text{Energy (ATP)}$$

Oxidation (loss of H^+ ions)

Reduction (gain of H^+ ions)

Fossile Fuels:-

(1) Fossil fuels (oil, gas, coal) formed ~~an~~ from ancient aquatic life under sediment layers.

(2) petroleum and natural gas face depletion issues, with estimates suggesting petroleum may be 80% depleted by 2030.

(3) Coal reserves can last centuries, prompting research into converting coal into liquid and gaseous fuels for sustainability.

Rocket fuels!

⇒ Rocket propellant is the reaction mass of a rocket, this reaction mass is ejected at the highest achievable velocity from a rocket engine to produce thrust. Liquid hydrogen is used a a fuel for rocket propellant.

Nuclear Fission:-

⇒ It in a reaction where the nucleus of an atom splits into two or more smaller nuclei, while releasing Energy

$$^{236}_{92}U + ^{1}_{0}n = ^{236}_{92}U$$

$$^{236}_{92}U = ^{144}_{56}Ba + ^{89}_{36}Kr + 3\,^{1}_{0}n + 177\,MeV.$$

Nuclear Fusion:-

⇒ Nuclear Fusion in a reaction in which two or more atomic nuclei, usally Deuterium and tritium combine to form one or more different atomic nuclei and subatomic particles.

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$^{3}_{1}$Deuterium + $^{3}_{1}$tritium = $^{4}_{2}He + ^{1}_{0}n + 17.6 MeV$

Nuclear Fission Vs Fusion:-

Nuclear Fission	Nuclear Fusion
⇒ A heavy Nucleus breaks up to form two lighter nuclei.	⇒ Two light nuclei combine to form a heavy nucleus.
⇒ It involves a chain reaction	⇒ Chain reaction is not involved.
⇒ The heavy nucleus is bombarded with neutrons	⇒ light nuclei are heated to an extremely high temp.
⇒ We have proper control on that reaction	⇒ We have not proper control on that reaction.
⇒ Disposal of nuclear waste in a great environmental problem.	⇒ Disposal of nuclear waste in not involved.
⇒ Raw materials in not easily available and costly	⇒ Raw materials is easily avaible and cheaper.

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#Phase Rule & Phase Diagram:

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#phase - rule:-

⇒ phase - rule is a general principle governing "PVT" Systems.

$$\boxed{F = C - P + 2}$$

F = degree of freedom.

C = Components (no. of components in a system)

P = number of phase.

#Characteristics of phase rule:-

1. It deals with the behavior of heterogeneous System.

2. It predicts qualitive effects of pressure, temperature and Concentration changes on a heterogeneous equilibrium system diagrams.

3. It allows us to predit the number of stable phases that may exist in equilibrium for a particular system.

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Homogeneous System:-

⇒ A system consisting of one phase is called a homogeneous system.

Heterogeneous System:-

⇒ A system consisting two or more phases is called heterogenous system.

Phase: (P)

⇒ It can refer to the physical state of matter, such as solid, liquid, gas, or plasma.

Ex:-

1. A system containing liquid water in 1 - phase system.
2. A system containing liquid water and water vapor is 2- Phase system.
3. A system containing liquid water, water vapor, and solid ice in 3- phase system.

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More Ex:-

(a) pure substance (solid, liquid, gas): O_2, C_2H_6, H_2O → 1-phases.

(b) Mixtures of gases: O_2 and N_2 → 1-phase system.

(c) Miscible liquids: H_2O and ethanol → 1 phase system.

(d) Non-miscible liquids: $CHCl_3$ and H_2O → 2-phase system.

(e) Aqueous solution: NaCl in H_2O → 1-phase system.

(d) Mixtures of solids: monoclinic and rhombic sulfur → 2-phase system.

Components (C)

⇒ The number of components (C) is the number of chemically independent constituents of the system.

Ex:-

(1) H_2O and sulfur system → 1-component system.

here, H_2O = 3-phases

Sulfur = 4-phases.

(2) Mixture of gases: (O_2 & N_2) → 1 phase; 2-components.

(3) NaCl solution: (H_2O & NaCl) → 1 phase; 2-components.

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Degree of Freedom (F)

⇒ It is the number of independent intensive variables.

→ A system $F=0$, non-variant → No degree of freedom

→ A system $F=1$, uni-variant → 1-degree of freedom.

→ A system $F=2$, bi-variant → 2-degree of freedom.

Ex:-

(i) For pure gas, $F=2$

(ii) For mixture of gas, $F=3$

(iii) For water ⟷ water vapor, $F=1$

One-Component system: Phase diagrams:-

⇒ For,

1-Component system, $C=1$

We know:-

$$F = C - P + 2$$

$$= 3 - P \quad \cdots\cdots (1)$$

Case 1:-

For, 1-phase:-

$$P = 1$$

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$\therefore F = 3 - 1 = 2$

$\therefore$ The System is bi-variant.

→ A single phase system represented by an 'Area' on a P, T-graph.

Case-2:- For, 2-Phase:-

$P = 2$

$\therefore F = 3 - 2 = 1$

$\therefore$ The system is mono-variant/Uni-variant

→ A two-phase system is depicted by a 'Line' on a P, T-graph.

Case-3:- For, 3-Phase:-

$P = 3$

$\therefore F = 3 - 3 = 0$

$\therefore$ The system is non-variant/In-varient

→ A three phase system is represented by 'Point' on P, T-graph.

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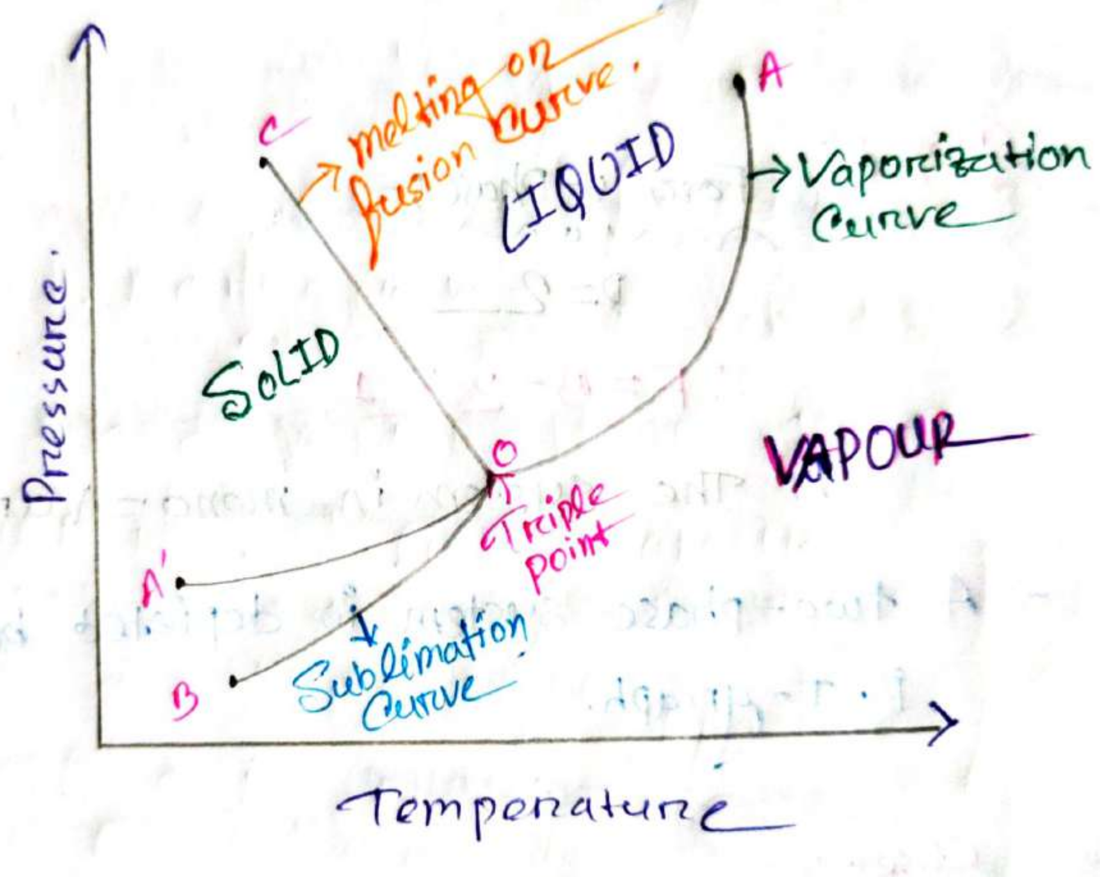
#phase-Diagrams:-

⇒ It is a graphical representation of the physical states of a substance under different conditions of temperature and pressure.



→ The diagram consists of

ⓐ The regions or areas.

ⓑ The lines or curves

ⓒ The triple point.

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Regions or areas:-

⇒ The diagram is divided into three areas

i.e. Solid, liquid, vapor.

⇒ For 1-phase and 1-component:-

$C = 1$
$P = 1$

$\therefore F = C - P + 2$
$= 1 - 1 + 2$

Thus each area of phase diagram represents a "bi-variant" system.

Line or Curves:-

⇒ There are three curve/lines separating the areas, showing the condition of equilibrium between any two phases.

(i) Melting or Fusion Curve:-

⇒ Solid and liquid phases are in equilibrium,

Solid $\rightleftharpoons$ Liquid ;

⇒ The Solid and liquid line is known as melting curve.

ii) Vaporisation Curve:-

⇒ Liquid and Vapour phases are in equilibrium, Liquid ⇌ Vapour;

→ Liquid and Vapour line is known as Vaporisation Curve.

iii) Sublimantion Curve:-

⇒ Solid and Vapour phases are in equilibrium, Solid ⇌ Vapour

→ Solid and Vapour line is known as Sublimantion Curve.

For two-phase and one-component:-

$C = 2 - 1 = 1$
$P = 2$

$\therefore$ $F = C - P + 2$
$= 1 - 2 + 2$
$= 1$

Thus each line of phase diagram represent a 'mono-variant' system.

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Triple point:-

⇒ The three boundary lines enclosing the three areas on the phase diagram intersect at a common point called Triple point.

A triple point shows the conditions under which all three phases (solid, liquid and vapor) can coexist in equilibrium.

Solid ⇌ Liquid ⇌ Vapor.

For:-

three phase, one component

$P = 3$

$C = 1$

$\therefore F = C - P + 2$

$= 1 - 3 + 2$

$= 0$

Thus at the triple point of phase diagram the system is 'non-variant'.

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Sulfur System:-

⇒ It is 'one - Component, four - phase System'.

The four phases are.

1) Two Solid Polymorphic forms

(i) Rhombic Sulfur (SR)

(ii) Monoclinic Sulfur (SM)

2) Sulfur Liquid (SL)

3) Sulfur Vapour (SV)

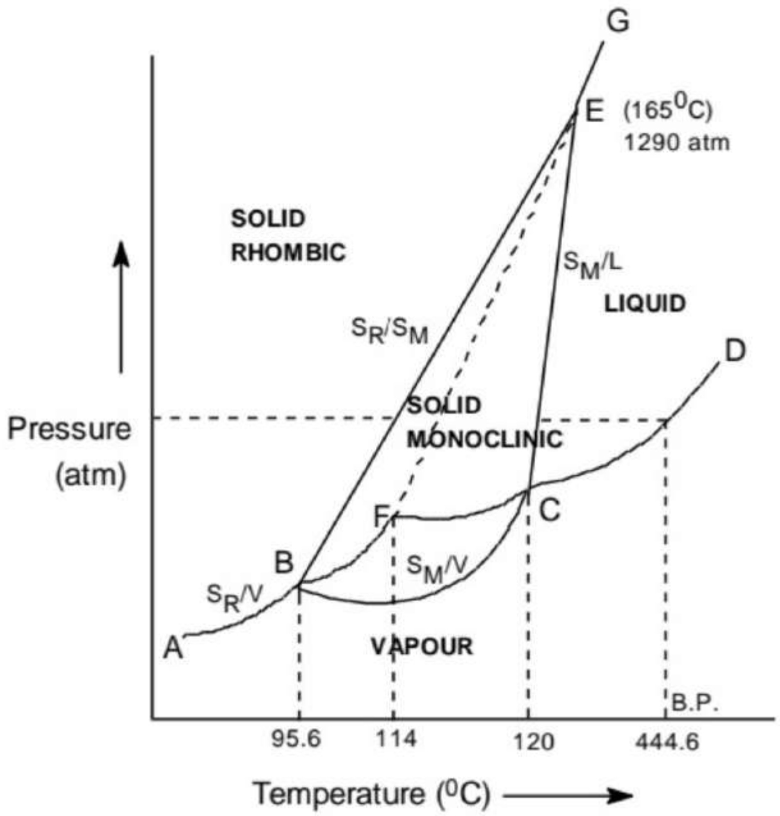

Fig. The phase diagram of the system 'Sulfur'

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(1)

The Areas of Sulfur System:-

⇒ It has one-phase and one component

$\therefore$ $p = 1, c = 1$

$F = c - p + 2$
$= 1 - 1 + 2$
$= 2$

two-degree of freedom 'bi-variant'

The Triple points of Sulfur system:-

⇒ It has three-phase and one component

$\therefore$ $p = 3$
$c = 1$

So, $F = c - p + 2$
$= 1 - 3 + 2$
$= 0$

no-degree of freedom 'non-variant'

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The Curves of Sulfur System:-

⇒ It has two-phase and one component

$\therefore P = 2$

$C = 1$

$\therefore F = C - P + 2$

$= 1 - 2 + 2$

$= 1$

One degree of freedom 'mono-variant'

→ The curves AB, BC, CD, CE, EG, BE

1. Curve AB, the vapor pressure curve of S_R
2. Curve BC, the vapor pressure curve of S_m
3. Curve CD, the vapor pressure curve of S_L
4. Curve BE, the transition curve, $S_R + \varphi \rightarrow S_m$
5. Curve CE, the fusion curve of S_m
6. Curve EG, The fusion curve of S_R

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